

Chien Nguyen

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Research Summary

My research focuses on **efficient and scalable generative modeling** for Large Foundation Models. I develop **novel model architectures, training paradigms, and inference algorithms** to improve computational efficiency, long-context modeling, and reasoning capabilities. More broadly, I aim to advance foundation models that **scale effectively with compute and data** across training and deployment.

Education

University of Oregon, USA

PhD in Computer Science, advised by Prof. Thien Huu Nguyen

Sep 2023 – May 2027 (Expected)

Hanoi University of Science and Technology, Vietnam

BSc in Computer Science

Sep 2018 – May 2023

Experience

Amazon Web Services (AWS)

Applied Scientist Intern

Santa Clara, USA

Jun 2026 – Present

- Developing speculative decoding methods for efficient LLM inference.

Google DeepMind

Student Researcher

Mountain View, USA

Sep 2025 – Jan 2026

- Improved LLM inference efficiency by enabling cross-model KV-cache compatibility via KV-cache fusion.

Adobe Research

Research Scientist Intern

San Jose, USA

Jun 2024 – Sep 2025

- Developed efficient architectures and training methods for long-context LLMs and small multilingual language models under limited-data constraints.

VinAI Research (now Qualcomm AI Research)

AI Research Resident

Hanoi, Vietnam

Mar 2022 – Sep 2023

- Researched structured information extraction and multilingual large language models.

Selected Publications and Preprints

- 2026 Chien Van Nguyen, Chaitra Hegde, Van-Cuong Pham, Ryan A. Rossi, Franck Deroncourt, Thien Huu Nguyen. **Orthrus: Memory-Efficient Parallel Token Generation via Dual-View Diffusion.** *arXiv preprint arXiv:2605.12825*. [PDF](#)
- 2026 Huy Huu Nguyen, Chien Van Nguyen, Franck Deroncourt, Ryan A. Rossi, Linh Ngo Van, Jieyang Chen, Thien Huu Nguyen. **CRISP: Cliff-awaRe Input-adaptive Sparse Prefilling with Structural-Mass-Motivated Routing.** In *Proceedings of the 31st Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*. [PDF](#)
- 2026 Chien Van Nguyen, Ryan Rossi, Linh Ngo, Franck Deroncourt, Thien Nguyen. **Octopus: Gated Selective Attention for Memory-Bounded Long-Context Inference in Large Language Models.** In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*. [PDF](#)
- 2025 Chien Van Nguyen, Huy Nguyen, Ryan A. Rossi, Trung Bui, Nikos Vlassis, Franck Deroncourt, Thien Huu Nguyen. **Lizard: An Efficient Linearization Framework for Large Language Models.** In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*. [PDF](#)
- 2025 Chien Van Nguyen, Huy Huu Nguyen, Ryan A. Rossi, Trung Bui, Viet Lai, Franck Deroncourt, Thien Nguyen. **Taipan: Efficient and Expressive State Space Language Models with Selective Attention.** *arXiv preprint arXiv:2410.18572*. [PDF](#)

- 2024 Thuat Nguyen, Chien Van Nguyen, Viet Dac Lai, Hieu Man, Nghia Trung Ngo, Franck Dernoncourt, Ryan A. Rossi, Thien Huu Nguyen. **CulturaX: A Cleaned, Enormous, and Multilingual Dataset for LLMs in 167 Languages**. In *Proceedings of the 2024 International Conference on Computational Linguistics (COLING 2024)*. [PDF](#)
- 2023 Viet Lai*, Chien Van Nguyen*, Nghia Ngo, Thuat Nguyen, Franck Dernoncourt, Ryan A. Rossi, Thien Huu Nguyen. **Okapi: Instruction-tuned Large Language Models in Multiple Languages with Reinforcement Learning from Human Feedback**. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP 2023)*. [PDF](#)
- 2023 Chien Van Nguyen, Huy Huu Nguyen, Franck Dernoncourt, Thien Huu Nguyen. **Transitioning Representations between Languages for Cross-lingual Event Detection via Langevin Dynamics**. In *Findings of the Association for Computational Linguistics: EMNLP (EMNLP 2023)*. [PDF](#)
- 2023 Chien Van Nguyen, Linh Ngo Van, Thien Huu Nguyen. **Retrieving Context to Align Representations for Cross-lingual Event Detection**. In *Findings of the Association for Computational Linguistics: ACL (ACL 2023)*. [PDF](#)
- 2023 Chien Van Nguyen, Hieu Man Duc Trong, Thien Huu Nguyen. **Contextualized Soft Prompts for Extraction of Event Arguments**. In *Findings of the Association for Computational Linguistics: ACL (ACL 2023)*. [PDF](#)
- 2023 Huy Nguyen, Chien Van Nguyen, Linh Ngo, Luu Anh Tuan, Thien Huu Nguyen. **A Spectral Viewpoint on Continual Relation Extraction**. In *Findings of the Association for Computational Linguistics: EMNLP (EMNLP 2023)*. [PDF](#)

*Equal contribution.

Projects

- Orthrus: Dual-View Diffusion for Fast LLM Inference** [🔗 chiennv2000/orthrus](#) *Lead Author*
- Developed the first diffusion-based LLM framework for lossless generation via a novel dual-view architecture, enabling up to 7.8× faster inference with no quality degradation and robust long-context stability.
- Okapi: Multilingual LLM Training & Evaluation** [🔗 nlp-uoregon/Okapi](#) *Lead Author*
- Developed a multilingual instruction-tuning and evaluation framework supporting 26 languages, with its benchmarks adopted by major LLM organizations including Qwen (Alibaba), Stability AI, and Cohere.
- GPT-OSS Efficient Finetuning** [🔗 nlp-uoregon/oregon_gpt_oss_patching](#) *Lead Author*
- Implemented custom Triton-based flash attention kernels with a complete backward pass for OpenAI's GPT-OSS models (20B and 120B), enable memory-efficient training on long contexts with a 2.5× throughput gain.
- Vistral-7B-Chat** [🔗 Viet-Mistral/Vistral-7B-Chat](#) *Lead Author*
- Built the first open-source Vietnamese LLM, achieving SOTA performance on Vietnamese benchmarks at release and outperforming ChatGPT and Gemini; **200,000+ downloads** on HuggingFace.

Technologies

- Languages & Frameworks:** C/C++, PyTorch, JAX/Flax, CUDA, Triton, CuTeDSL
- Libraries & Tools:** Git, Bash, SLURM, HuggingFace, vLLM, SGLang, TensorRT-LLM

Academic Services

Conference Reviewer: ACL, EMNLP, NAACL, COLM, AACL, ICLR, NeurIPS

Awards

- Gurdeep Pall Graduate Student Fellowship** – *University of Oregon 2026*
Awarded to outstanding graduate students for demonstrating exceptional academic and research excellence.
- Lokey Graduate Science Award** – *University of Oregon 2023-24*
Prestigious university-wide recognition granted to top-tier graduate students in science fields.
- First Prize** – *BKAI-NAVER Hackathon Challenge 2022*
Ranked 1st among 80 teams in Intent Detection and Slot Filling.
- First Prize** – *SoICT-IBM Hackathon 2020*
Ranked 1st out of 20 finalist projects for developing an open-domain chatbot.